

PATHAHEAD · FREE ROADMAP

Become a Cloud / DevOps Engineer from Scratch

The engineers who ship and run software at scale. Go from zero to job-ready in about five months — with real pipelines, containers, and cloud deployments.

22

weeks · ~10–12 hrs/wk

5

hands-on projects

4

phases · to production

₹0

free tiers only

A free PathAhead learning resource · adjust the pace to your life — the order matters more than the dates.

THE ONE RULE

Automate → containerize → deploy

DevOps is learned by building pipelines, not reading about them. Everything below runs on free cloud tiers and ends in something deployed and automated.

THE ROADMAP

Four phases, step by step

0

Foundations

Weeks 1–5

The Linux, Git, and scripting base everything sits on.

WEEK 1

Linux & CLI

Shell, files, permissions, processes.

WEEK 2

Networking basics

DNS, HTTP, ports, load balancing.

WEEK 3

Git workflows

Branches, PRs, merge strategies.

WEEK 4

Scripting

Bash and Python for automation.

WEEK 5

Programming refresher

Enough code to build tooling.

1

Containers & CI/CD

Weeks 6–11

Package apps and ship them automatically.

WEEK 6

Docker

Images, containers, Dockerfiles.

WEEK 7

Docker Compose

Multi-service local environments.

WEEK 8

CI/CD concepts

Pipelines with GitHub Actions.

WEEK 9

Build & deploy

Automated test-and-deploy workflows.

WEEK 10

Registries

Push and pull container images.

WEEK 11 · PROJECT ►

Containerized pipeline

Dockerize an app and auto-deploy it.

2

Cloud & Orchestration

Weeks 12–17

Run workloads on AWS and Kubernetes.

WEEK 12**Cloud fundamentals**

AWS core: EC2, S3, IAM.

WEEK 13**Cloud networking**

VPCs, subnets, security groups.

WEEK 14**Kubernetes basics**

Pods, deployments, services.

WEEK 15**K8s scaling**

Replicas, autoscaling, config.

WEEK 16**Infrastructure as Code**

Terraform to define infrastructure.

WEEK 17 · PROJECT ►**Deploy to Kubernetes**

Run your app on a K8s cluster.

3

Monitoring & SRE

Weeks 18–22

Keep production healthy and secure.

WEEK 18**Monitoring**

Prometheus and Grafana dashboards.

WEEK 19**Logging**

Centralized logs with the ELK stack.

WEEK 20**Security & secrets**

Secrets management, least privilege.

WEEK 21**Cert prep**

AWS Cloud Practitioner study.

WEEK 22 · CAPSTONE ►**Full platform**

CI/CD + IaC + K8s + monitoring, end to end.

BUILD A PORTFOLIO

Five projects that get you hired

Projects beat certificates. Do them in order — each builds on the last. Put every one on GitHub with a clear README: problem → approach → result.

Start here

1 • Dockerize an App

Take any app and package it into a container that runs anywhere.

CI/CD

2 • CI/CD Pipeline

GitHub Actions that test and deploy on every push.

Cloud

3 • Deploy to AWS

Host an app on EC2/S3 with proper IAM and networking.

In demand

4 • Kubernetes Deployment

Run a multi-service app on a cluster with scaling.

Capstone

5 • IaC + K8s + Monitoring

A fully automated, monitored platform defined in Terraform.

ROLES YOU CAN APPLY FOR

DevOps Engineer · Cloud Engineer · Site Reliability Engineer · Platform Engineer

TYPICAL STARTING SALARY · INDIA

₹6–14 LPA

LEARN IT

Learning resources

Every item links to the real source (clickable in this PDF).

BOOKS

- [The Phoenix Project](#) by Gene Kim, Kevin Behr and George Spafford
- [The DevOps Handbook](#) by Gene Kim, Jez Humble, Patrick Debois and John Willis
- [Site Reliability Engineering](#) by Google

COURSES

- [AWS Cloud Practitioner Essentials](#) by AWS (free)
- [DevOps on AWS](#) by Coursera
- [Docker and Kubernetes](#) by freeCodeCamp (free)
- [Google Cloud Fundamentals](#) by Coursera

YOUTUBE

- [TechWorld with Nana](#)
- [freeCodeCamp.org](#)
- [KodeKloud](#)

PRACTICE

- [KodeKloud](#)
- [AWS Free Tier](#)
- [GitHub](#)

DO THIS TODAY

Your first step (takes 30 minutes)

1

Install Docker Desktop

Free — your local container playground.

2

Open an AWS free-tier account

aws.amazon.com/free — enough to learn the core services.

3

Run one container

`docker run hello-world`, then containerize a small app. Started.

Job-ready checklist: 3–5 solid projects on GitHub (≥ 1 deployed or documented) · can explain your work in an interview · comfortable with the core tools of the field · a portfolio or profile recruiters can find.

PathAhead — India's premium career guidance platform. This roadmap is a free educational resource; timelines are a guide and everyone learns at a different pace. Referenced third-party resources (KodeKloud, freeCodeCamp, AWS Skill Builder, TechWorld with Nana (YouTube), Docker docs) are for learning only and are not affiliated with PathAhead.